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# Interpretable Control for Unstable Systems via SINDy-RL

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# What if the controller was just an equation?

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$$u(x) = \underbrace{-2.4 \theta_1}_{\text{balance}} - \underbrace{0.9 \theta_2}_{\text{balance}} - \underbrace{0.5 \dot{\theta}_1}_{\text{damp}} - \underbrace{0.2 \dot{\theta}_2}_{\text{damp}} + \dots$$

~200 terms · 47× fewer parameters than the NN · most coefficients have physical interpretation

# The Inverted Double Pendulum

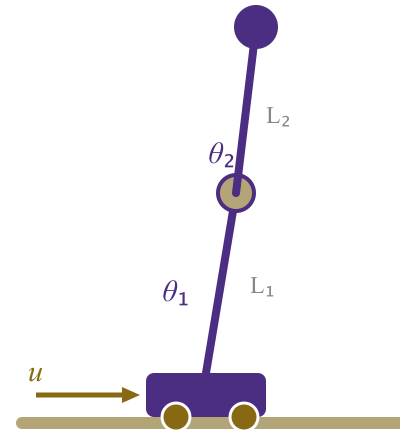
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**State**  $x \in \mathbb{R}^6$

$$x = \begin{bmatrix} x_{\text{cart}} \\ \theta_1 \\ \theta_2 \\ \dot{x}_{\text{cart}} \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

**Action**  $u \in [-1, 1]$  — cart force

**Unstable equilibrium** at  $\theta_1 = \theta_2 = 0$



$L_1 = L_2 = 0.6 \text{ m}$  ·  $dt = 0.05 \text{ s}$  · MuJoCo physics

# SINDy — Discovering equations from data

**Key idea:** dynamics live in a low-dimensional function space.

$$\dot{X} = \underbrace{\Theta(X, U)}_{\text{candidate feature library}} \cdot \underbrace{\Xi}_{\text{sparse coefficients}}$$

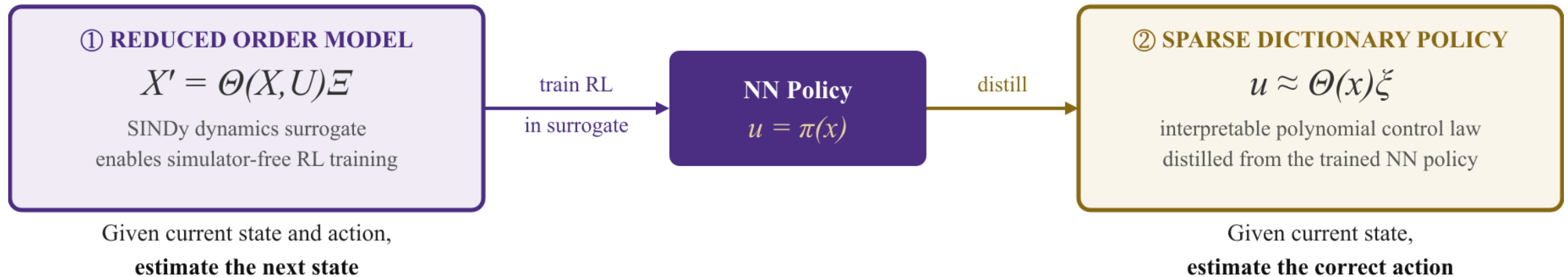
$$\Theta = \left[ 1 \mid x \mid \theta_1, \theta_2 \mid x^2, x\theta_1, \theta_1^2, \dots \right] \quad \text{degree-}d \text{ polynomial library}$$

**STLSQ solver** drives most coefficients to *exactly zero* — leaving a sparse equation with only the dominant governing terms.

**Library degree  $d$  is a design choice** — it must match the complexity of the system and is not obvious a priori.

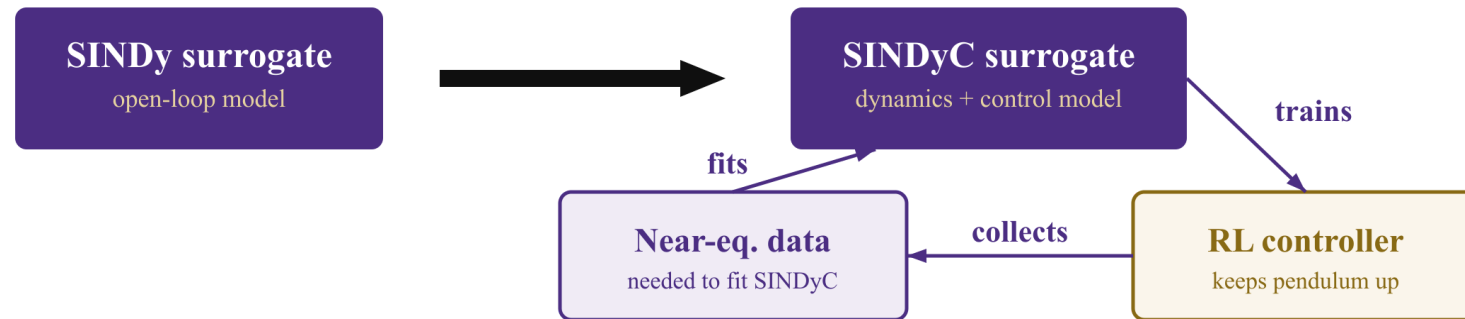
# Two objectives

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# SINDyC is a chicken-and-egg problem

You can't train near equilibrium without reaching it — and you can't reach it without a controller.



Solution: co-train the controller and surrogate in an iterative active-learning loop.

# Baseline — the performance ceiling

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A standard PPO agent trained with **full simulator access**.

**100%**

Task success rate  
(20/20 eval episodes)

**9359**

Mean reward  
(max possible  $\approx 10,000$ )

**9,731**

Policy parameters  
MLP  $[64 \times 64]$

# Sparse policy distillation — the compounding error trap

**Behavioral cloning** on 50,000 oracle transitions:

$$\min_{\Xi} \left\| \Theta(X) \Xi - U^* \right\|_2 + \lambda \|\Xi\|_1$$

210-term degree-4 library → STLSQ retains  
**206 terms · 47× fewer params** than NN

**But the policy fails under noise:**

| Noise $\sigma$ | Mean episode length |
|----------------|---------------------|
| 0 (training)   | ~1000 steps ✓       |
| 0.1            | ~200 steps          |
| 0.3            | ~20 steps ✗         |

Off-distribution states produce small errors  
→ errors compound → catastrophic failure.



# The fix: query the oracle, for free

**Key insight:** the NN policy is a pure function of state — no memory, no rollouts needed. Query it at *any* state we choose.

For each of 3 rounds:

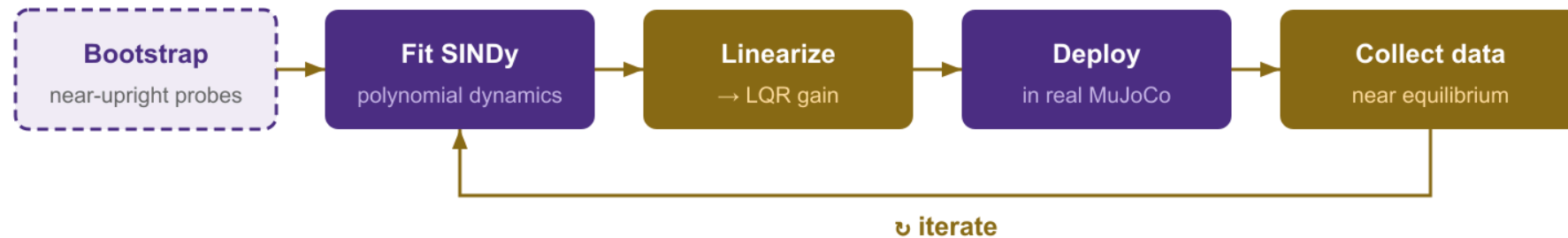
1. Perturb states:  $\tilde{x} \sim \mathcal{N}(x, \sigma^2)$
2. Query oracle:  $\tilde{u} = \pi_{\text{NN}}(\tilde{x})$
3. Append  $(\tilde{x}, \tilde{u})$  to dataset

Dataset grows **4×** (50k  $\rightarrow$  200k pairs).  
STLSQ re-fit recovers cross-coupling terms.

|             | $\sigma = 0.1$ | $\sigma = 0.3$ |
|-------------|----------------|----------------|
| Base policy | ~200 steps     | ~20 steps ✗    |
| Augmented   | ~1000 steps ✓  | ~500–900 steps |

**25–45× more robust** — zero additional simulator interactions.

# ROM surrogate – SINDy dynamics + LQR control



## PPO on SINDy fails to transfer

Exploits polynomial approximation errors — actions that score well in the model don't generalize: **24 steps** average in MuJoCo.

## Why LQR transfers

Uses only the *Jacobian at the upright fixed point* — accurate near equilibrium in both model and real system. No approximation errors to exploit.

# ROM surrogate — results

**SINDy RMSE convergence** (fixed validation set):

| Iteration     | RMSE  | Cumulative transitions |
|---------------|-------|------------------------|
| 0 (bootstrap) | 0.188 | 5,000                  |
| 1             | 0.182 | 10,000                 |
| 2             | 0.184 | 15,000                 |

Baseline NN: **400,000** real-sim steps.  
SINDy-LQR: **15,000** — **27× more data-efficient.**

**LQR transfer — real MuJoCo evaluation**  
(10 episodes per iteration):

| Iteration | Mean return | Mean length | Success     |
|-----------|-------------|-------------|-------------|
| 0         | 9359.88     | 1,000       | <b>100%</b> |
| 1         | 9359.87     | 1,000       | <b>100%</b> |
| 2         | 9359.90     | 1,000       | <b>100%</b> |

LQR gain computed from SINDy linearized around upright.

Mean return matches the full-order baseline (9,359) exactly.

# How do they compare?

| Approach                  | Real-sim steps | Policy type            | Mean length        | Success | Interpretable |
|---------------------------|----------------|------------------------|--------------------|---------|---------------|
| Baseline NN               | 400,000        | NN (9,731 params)      | 1,000              | 100%    | ✗             |
| Sparse policy (base)      | 400,000*       | Polynomial (206 terms) | ~20 @ $\sigma=0.3$ | Low     | ✓             |
| Sparse policy (augmented) | 400,000*       | Polynomial (206 terms) | ~500–900           | ~50–90% | ✓             |
| SINDy + PPO-in-surrogate  | ~15,000        | PPO (NN)               | ~24                | ~0%     | ✗             |
| SINDy-LQR                 | 15,000         | LQR from SINDy         | 1,000              | 100%    | ✓             |
| Phase 3 (stretch)         | TBD            | Polynomial             | —                  | —       | ✓             |

\* Inherits baseline training data — no additional agent training, one-shot regression from oracle queries.

# Stretch Goal

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## PHASE 3 · FULLY INTERPRETABLE CLOSED-LOOP CONTROL

Combine the interpretable **dynamics** (ROM) with the interpretable **policy** (sparse dictionary)

## Bonus Stretch Goal

### IF TIME PERMITS

Start from the down equilibrium position.

**Goal:** reproduce this with SINDy-RL — interpretable swing-up + interpretable stabilizer, end to end.



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# Interpretable control for unstable systems

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**THANK YOU**

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